

maxpack Benchmark Report

Generated 2026-03-27 20:08 · macOS-26.3-arm64-arm-64bit-Mach-O ·
Datasets: bat, cpython, fd, go, lsd

bat 53.28 MB

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
7z -mx=9	3.65 MB	14.6x	6.31s	977ms
maxpack	4.70 MB	11.3x	276ms	605ms
tar+xz	13.17 MB	4.0x	13.17s	3.40s
tar+bzip2	14.74 MB	3.6x	6.54s	4.38s
tar+zstd -3	16.15 MB	3.3x	3.08s	2.96s
tar+gzip	16.54 MB	3.2x	3.88s	3.00s
zip	18.57 MB	2.9x	1.96s	1.97s
tar+lz4	21.56 MB	2.5x	19.72s	10.12s

cpython 919.42 MB

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
maxpack	68.55 MB	13.4x	2.70s	0ms
7z -mx=9	90.99 MB	10.1x	93.34s	4.86s
tar+xz	192.24 MB	4.8x	234.99s	22.00s

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
tar+bzip2	222.49 MB	4.1x	64.73s	29.10s
tar+zstd -3	248.66 MB	3.7x	21.80s	18.48s
tar+gzip	262.43 MB	3.5x	39.08s	16.11s
zip	278.11 MB	3.3x	18.75s	10.59s
tar+lz4	385.51 MB	2.4x	17.41s	15.95s

fd

4.16 MB

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
7z -mx=9	141.7 KB	30.1x	338ms	51ms
tar+xz	156.7 KB	27.2x	629ms	166ms
maxpack	179.8 KB	23.7x	32ms	37ms
tar+zstd -3	341.3 KB	12.5x	166ms	160ms
tar+bzip2	710.1 KB	6.0x	391ms	212ms
tar+gzip	1.07 MB	3.9x	234ms	162ms
zip	1.14 MB	3.7x	74ms	88ms
tar+lz4	1.55 MB	2.7x	169ms	185ms

go 868.81 MB

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
maxpack	47.08 MB	18.5x	2.20s	7.61s
7z -mx=9	67.65 MB	12.8x	79.84s	11.81s
tar+xz	140.60 MB	6.2x	242.71s	42.94s
tar+bzip2	173.81 MB	5.0x	91.89s	49.82s
tar+zstd -3	190.81 MB	4.6x	43.36s	36.20s
tar+gzip	218.42 MB	4.0x	59.09s	37.30s
zip	255.12 MB	3.4x	27.91s	18.32s
tar+lz4	329.90 MB	2.6x	45.23s	36.52s

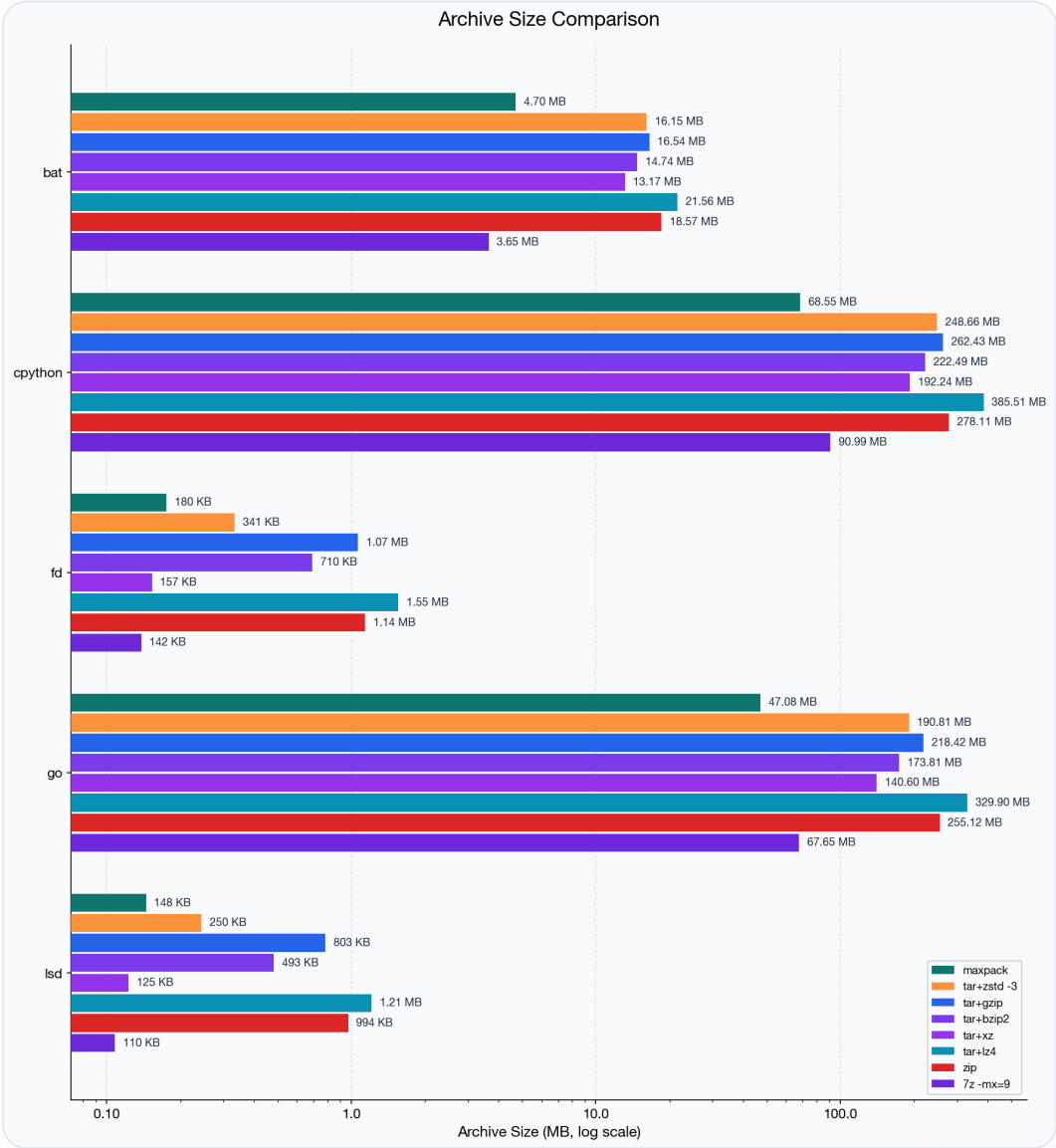
lsd 3.76 MB

METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
7z -mx=9	110.3 KB	34.9x	365ms	55ms
tar+xz	125.5 KB	30.7x	706ms	180ms
maxpack	148.2 KB	26.0x	29ms	40ms
tar+zstd -3	249.9 KB	15.4x	187ms	181ms
tar+bzip2	492.6 KB	7.8x	362ms	235ms
tar+gzip	802.9 KB	4.8x	238ms	173ms
zip	993.8 KB	3.9x	67ms	93ms

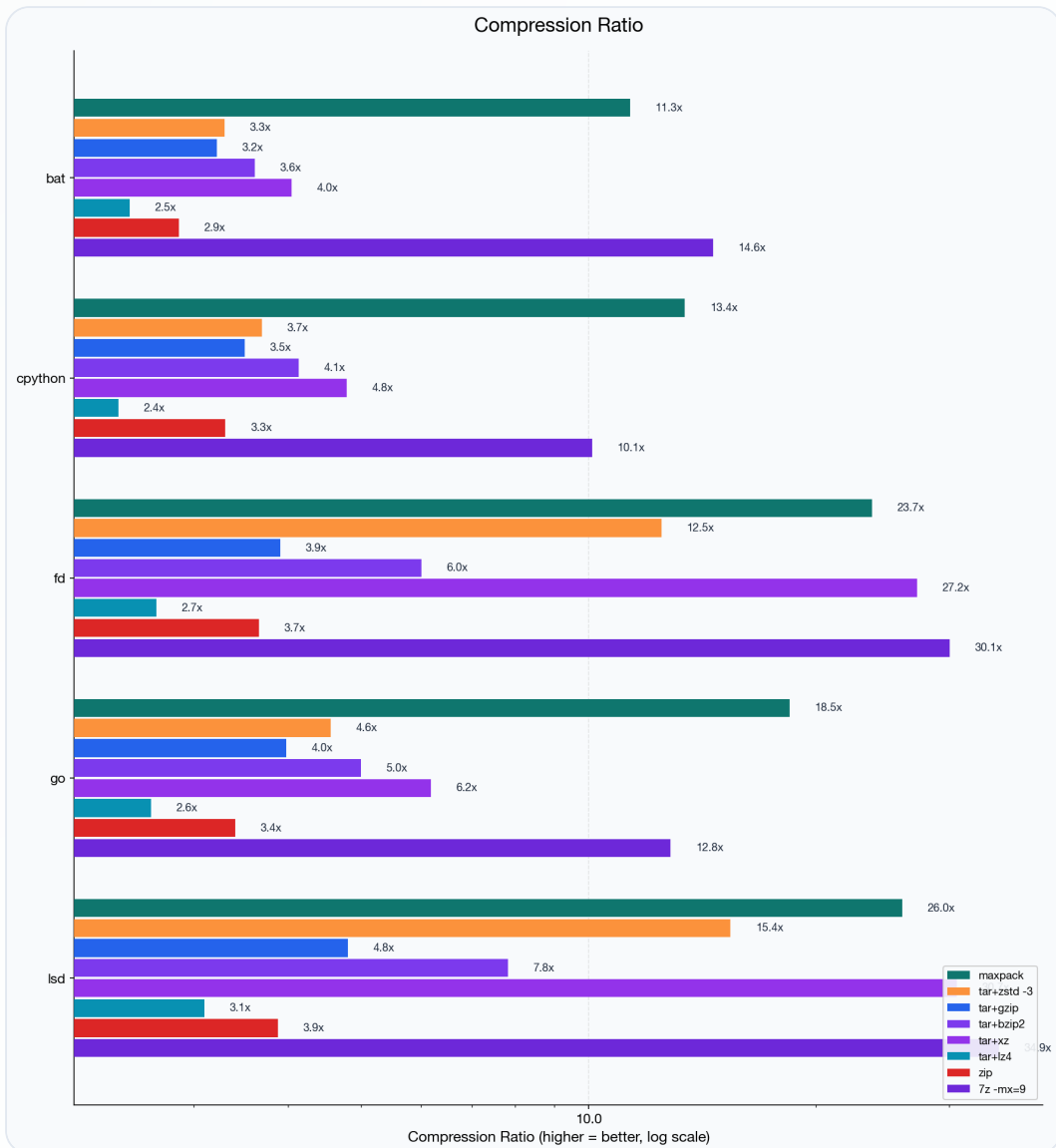
METHOD	ARCHIVE SIZE	RATIO	PACK	UNPACK
tar+lz4	1.21 MB	3.1x	184ms	176ms

Plots

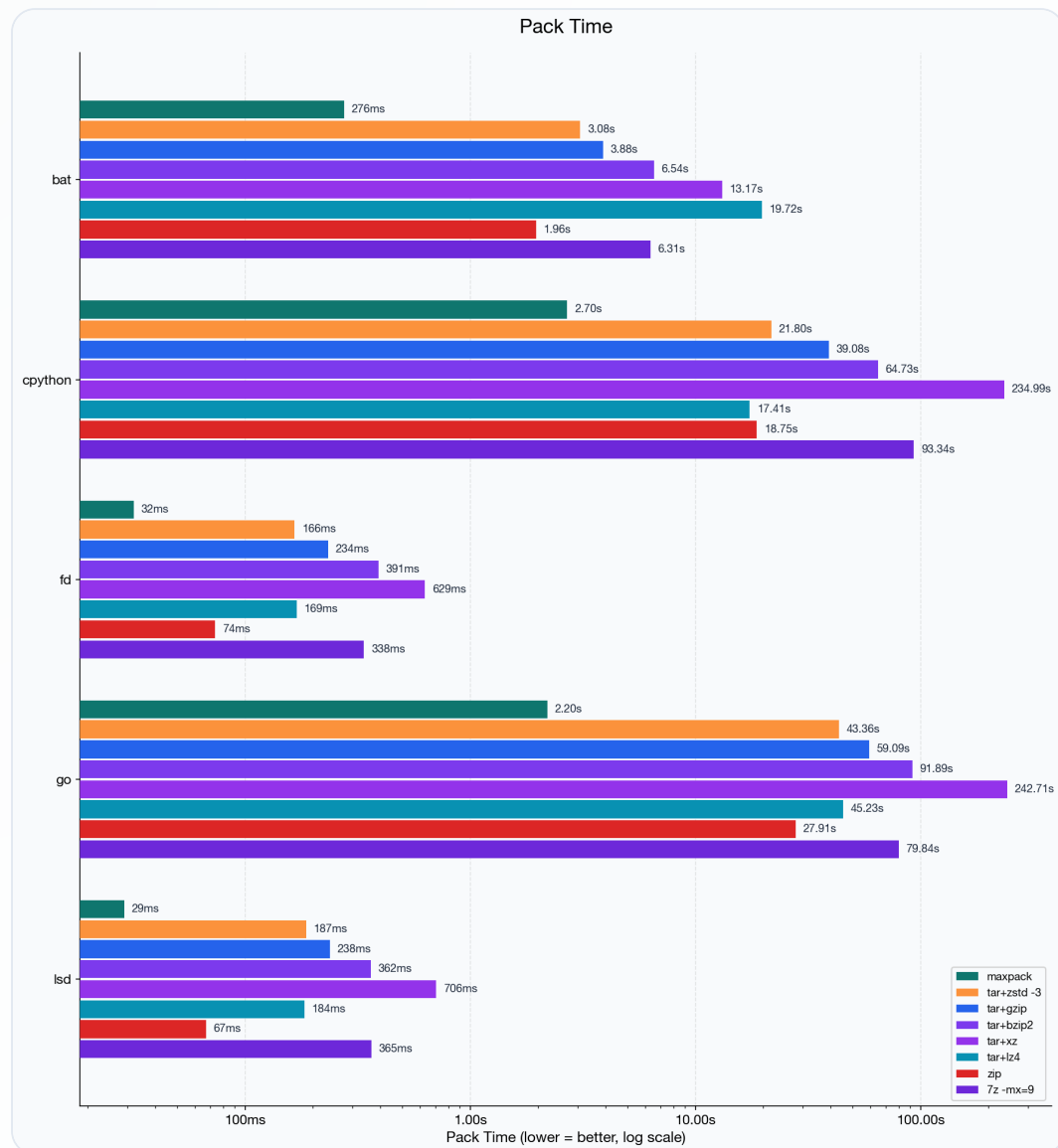
01 Archive Size



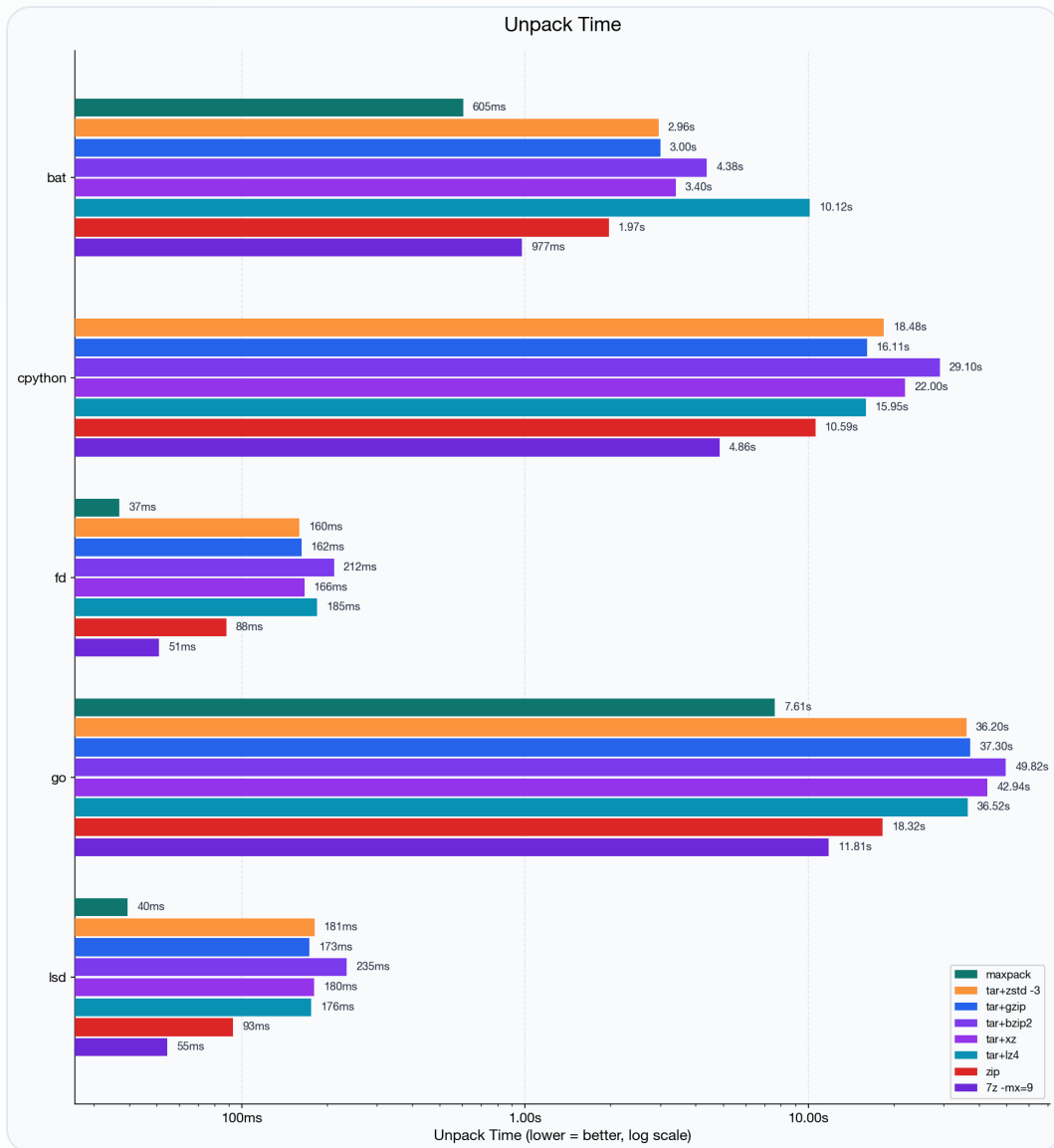
02 Compression Ratio



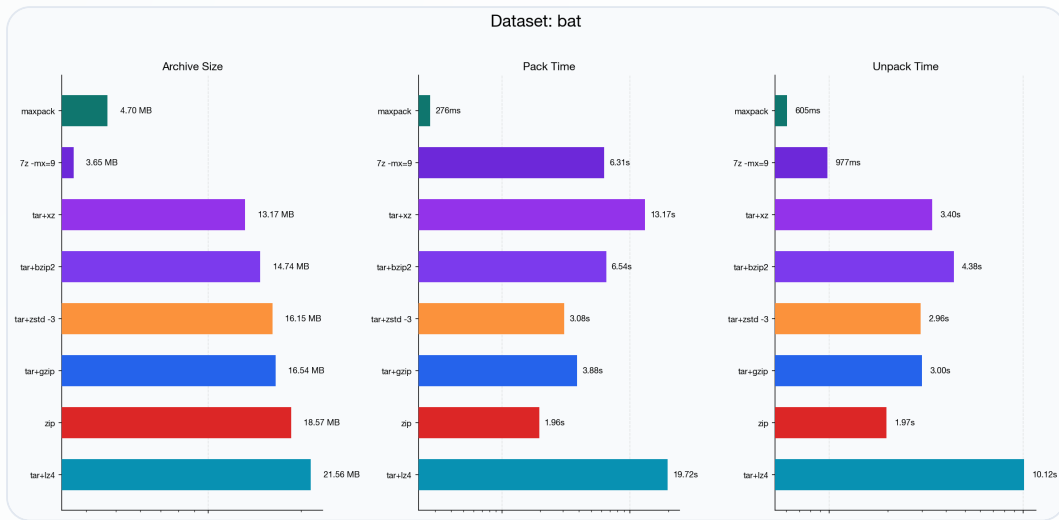
03 Pack Time



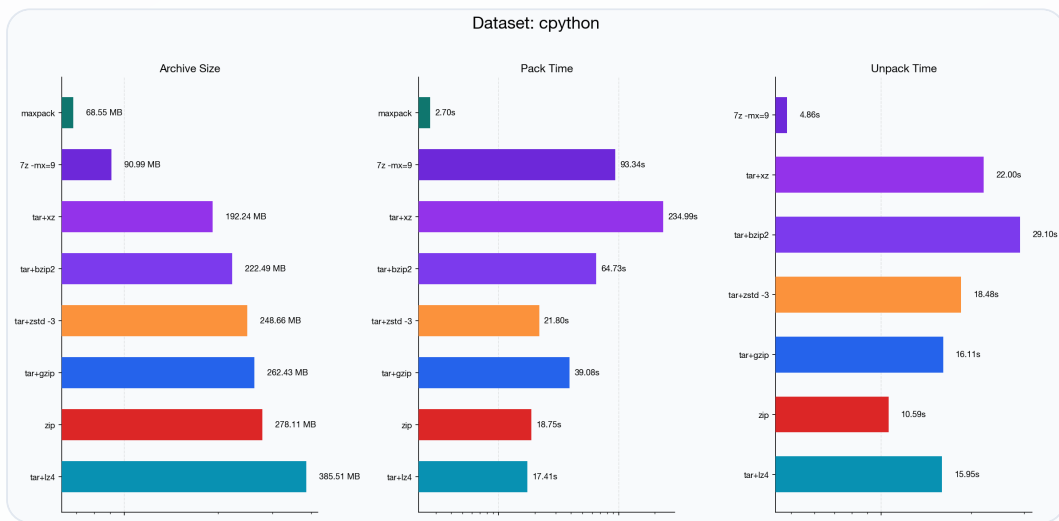
04 Unpack Time



05 Bat Detail



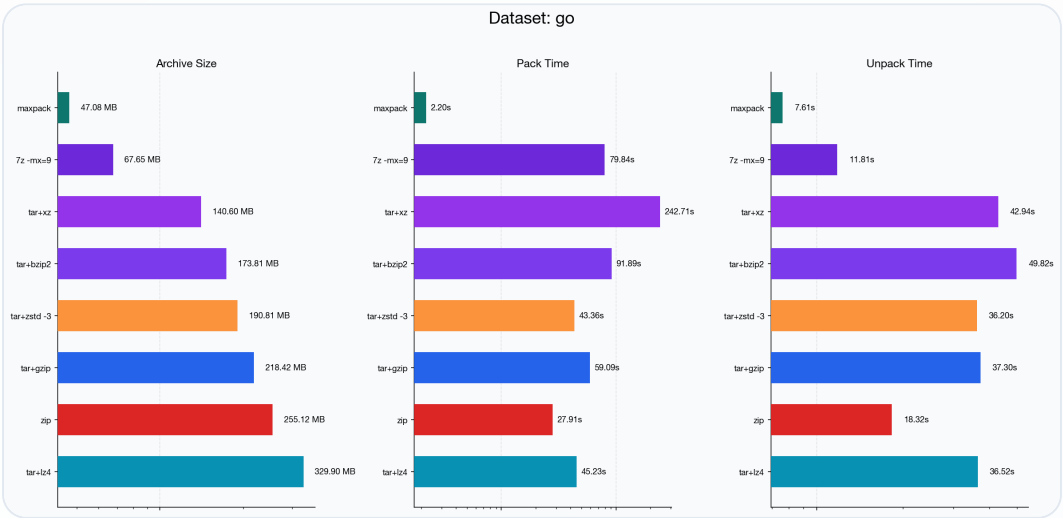
05 Cpython Detail



05 Fd Detail



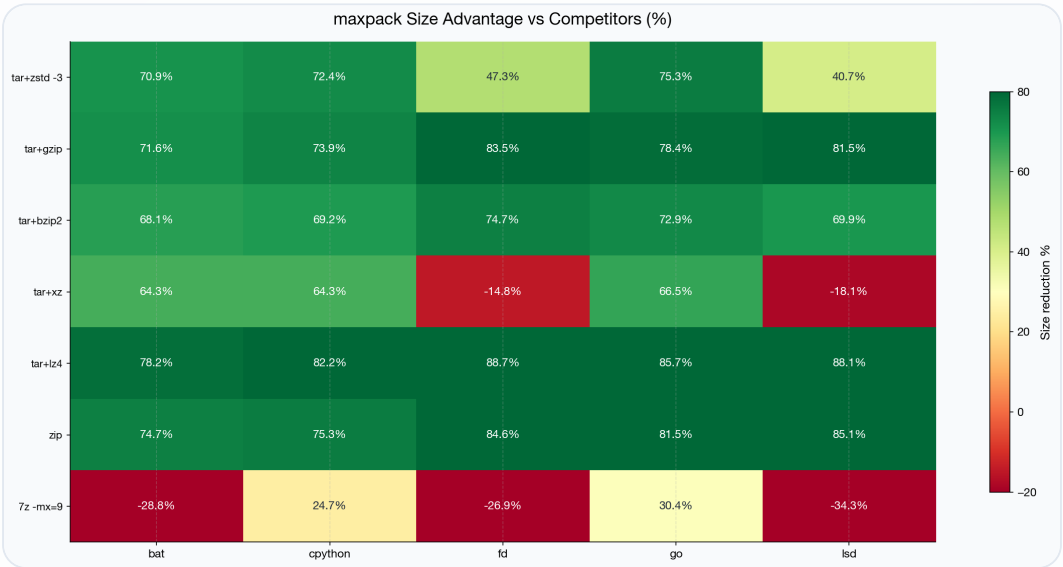
05 Go Detail



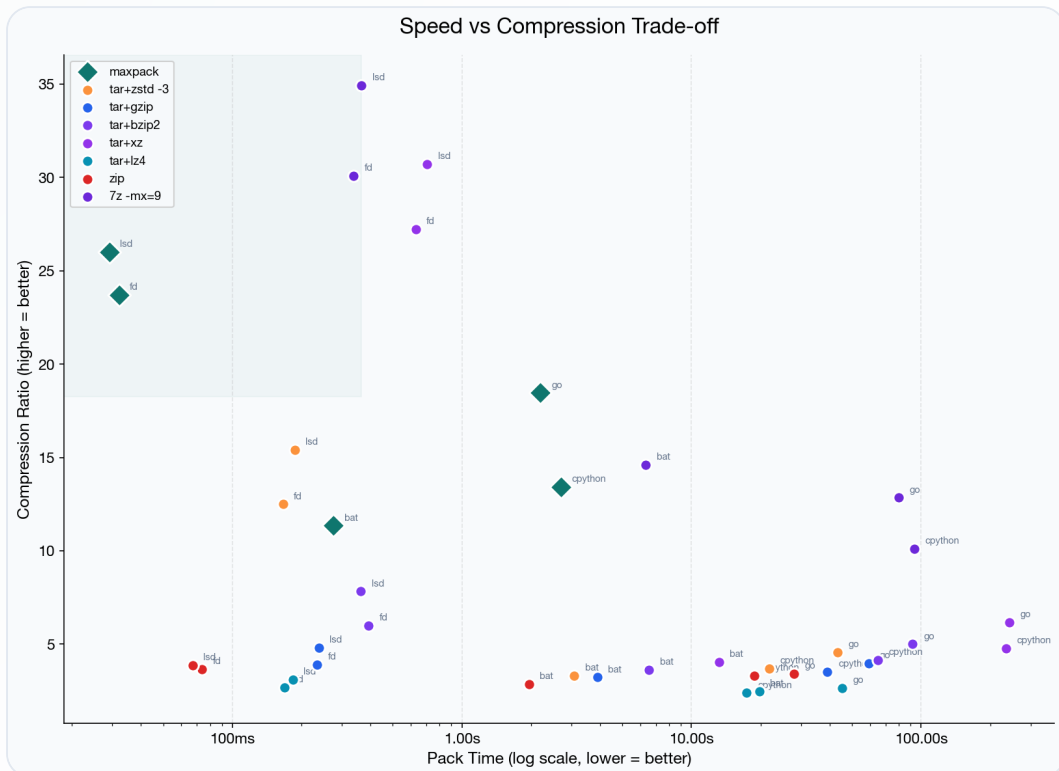
05 Lsd Detail



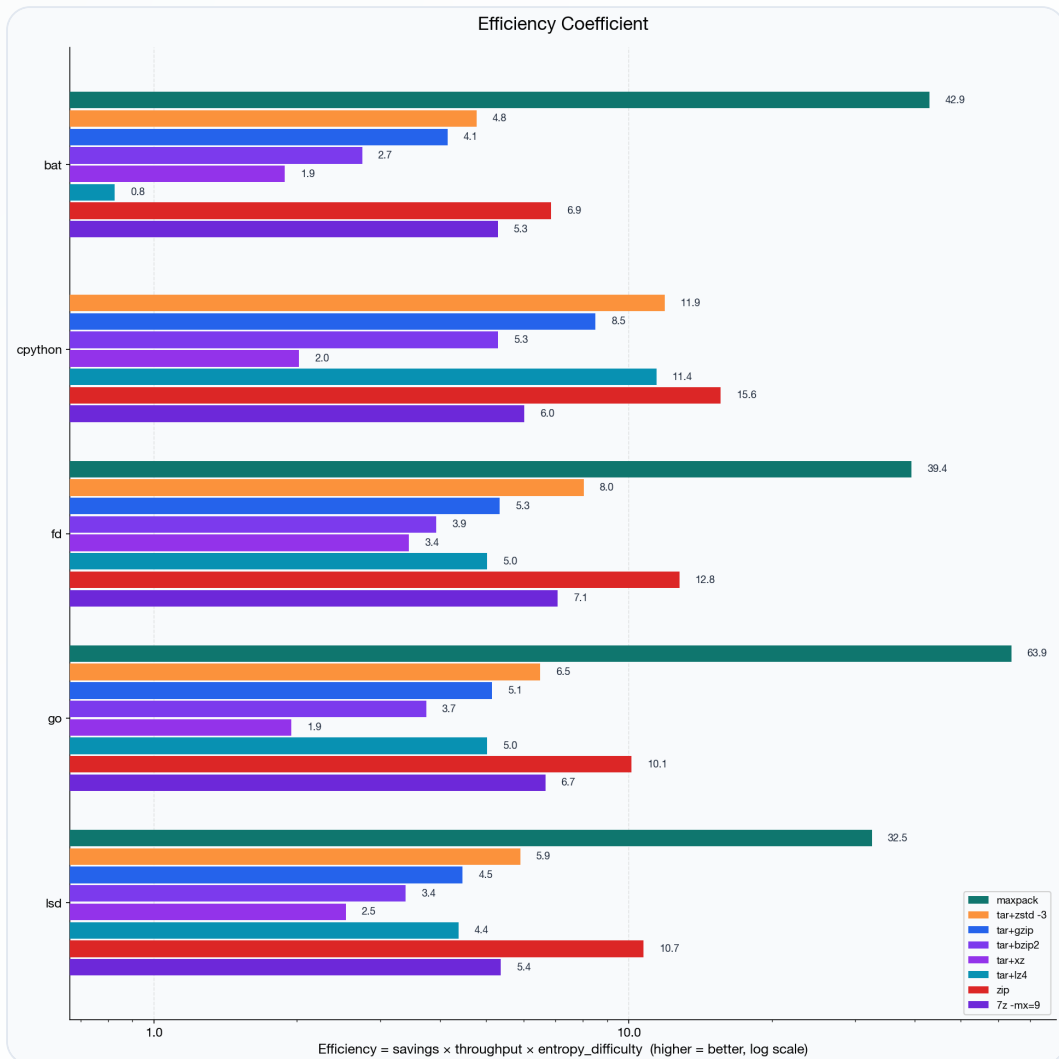
06 Advantage Heatmap



07 Speed Vs Ratio



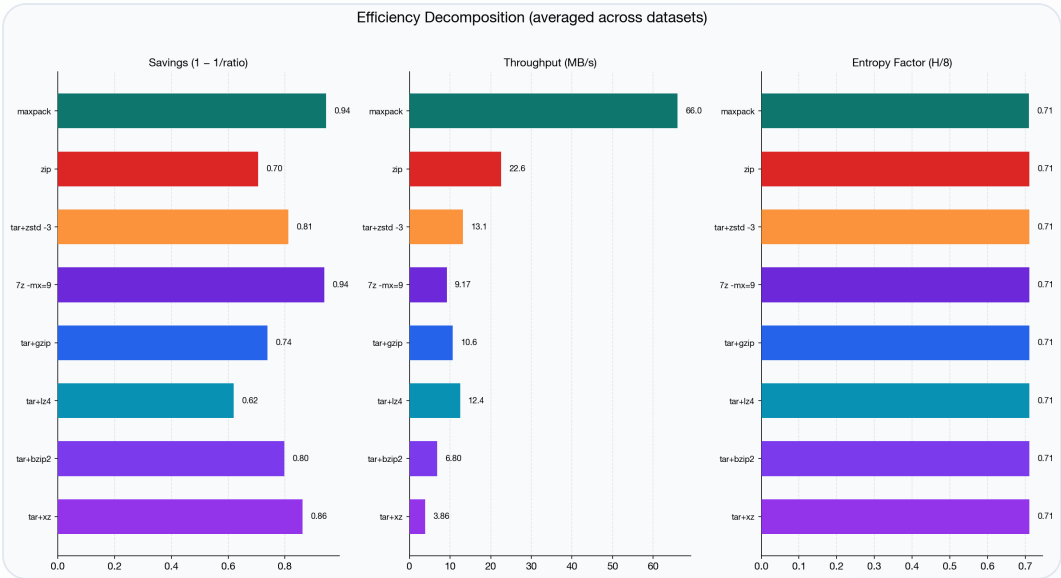
08 Efficiency



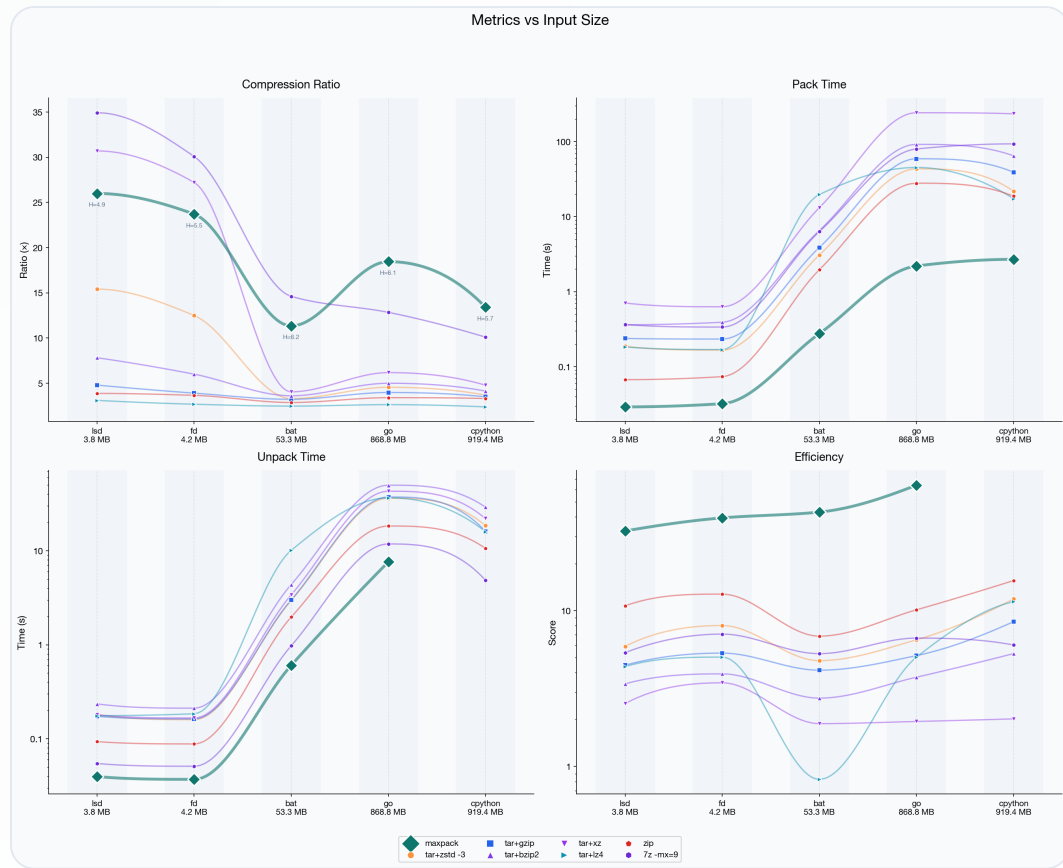
08B Efficiency Vs Entropy



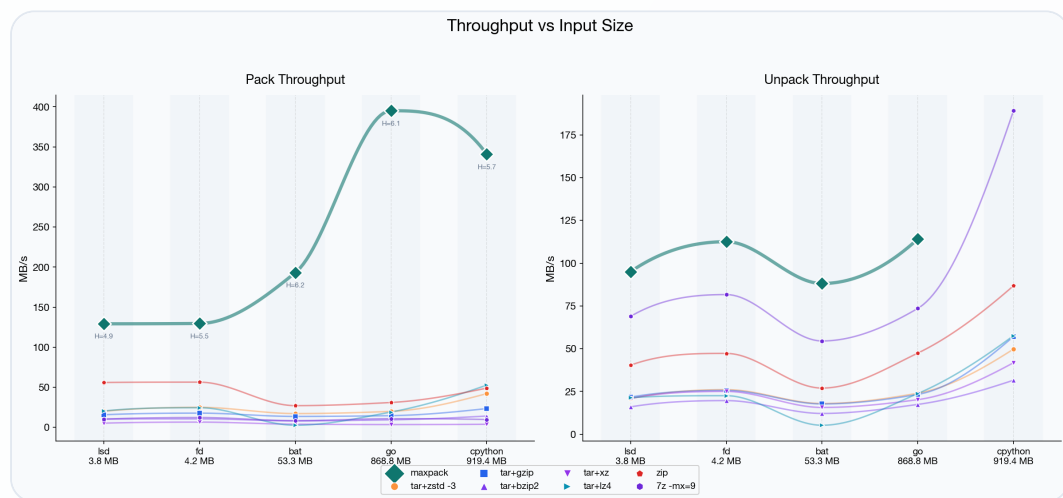
08C Efficiency Decomposed



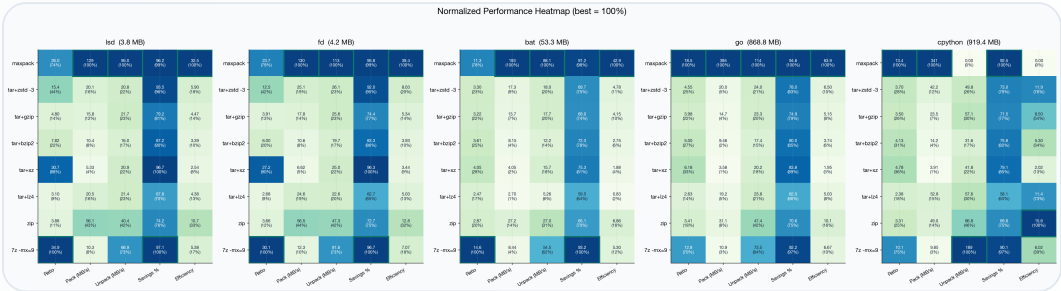
09 Curves Vs Size



09B Throughput Curves



09C Heatmap



09C Normalized Profiles

